Lab: Neural Architecture of Agents

Learning Goal: Map brain structures to their Active Inference functions.

Part 1: Brain Structure Mapping

Exercise: Complete the table mapping brain structures to Active Inference components.

Brain Structure	Active Inference Role	What happens if damaged?
Primary visual cortex (V1)		
Prefrontal cortex (PFC)		
Basal ganglia		
Thalamus		
Cerebellum		
Ventral tegmental area (VTA)		
Locus coeruleus		

Write your response here

Part 2: Tracing the Action Loop

Learning Goal: Follow the neural pathway from perception to action.

Scenario: You see a friend across the room and wave to them.

Trace the neural pathway through each structure, specifying what each does:

1. Visual cortex → What does it process?
2. Temporal cortex → What recognition occurs?
3. Prefrontal cortex → What decision is made?
4. Basal ganglia → How is the action selected?
5. Motor cortex → What command is issued?
6. Cerebellum → What prediction is generated?
7. Spinal cord / muscles → What happens?

Write your response here

Part 3: Neuromodulator Analysis

> **Learning Goal:** Understand how different neurotransmitters tune the system.

Case studies: For each drug/condition, identify which neuromodulator is affected and predict the Active Inference consequences.

1. **Caffeine** (increases noradrenaline): What happens to arousal and precision?
2. **Nicotine** (stimulates acetylcholine receptors): What happens to sensory attention?
3. **L-DOPA** (restores dopamine, used in Parkinson's): What improves? What might worsen?
4. **SSRI antidepressants** (increase serotonin): How might this affect temporal planning?

Write your response here

Part 4: The Cerebellum Challenge

> **Learning Goal:** Explore forward models through motor prediction.

Hands-on exercise:

1. Close your eyes and touch your nose with your index finger. Success? Your cerebellum's forward model is working.
2. Now try to write your name with your non-dominant hand. Notice the prediction errors — your forward model is less calibrated for this hand.
3. Why does practicing a motor skill (like typing) make it more automatic? Explain using cerebellar forward models.

Write your response here

Part 5: Reflection

In 150 words, reflect: The brain is not a single agent — it's a collection of structures working together. How does Active Inference unify these different structures under a single principled framework?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Structure-function mapping	Brain regions and Active Inference roles
2	Pathway tracing	Neural circuitry for perception-action
3	Pharmacological reasoning	Neuromodulators and precision
4	Motor self-observation	Cerebellar forward models
5	Integrative reflection	Active Inference as unifying framework